

Subandhu Agravanshi

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SUMMARY

Generative AI / Applied AI Engineer with **3.5+ years** building production agentic, RAG and ML systems across enterprise, fintech, gaming and media. Strong across **agentic AI, RAG, search and ranking, multimodal AI, fine-tuning, evaluation, classical ML, data engineering and backend systems**. Takes AI from experimentation to production: serving concurrent users at scale, cost and latency optimization, AI engineering and evals, and measurable business impact.

WORK EXPERIENCE

Pocket FM | *Generative AI Engineer* Jun 2026 - Present

- Architected and delivered an **organization-level knowledge search and analysis platform**. Built a **ReAct orchestrator** as the core, with **Text-to-SQL** and analytics sub-agents, **20+** tools and **12+** skills on intelligent routing, served through Claude Opus via **AWS Bedrock** and **LiteLLM** (Python/**OpenWebUI**, **Databricks SQL**). Used by **250+** concurrent users across domains at scale.
- Architected the organizational **agentic routing ontology**: clustered **120k** chats with **DBSCAN** into **5** domains and **30** categories. Retrieval used **cross-encoders** and **Gemini embeddings** via **Qdrant** for knowledge, intelligent expansion and KB routing, 2-pass filtering with **Cohere reranker** and aggregated scoring. Retrieval recall and accuracy **+40%**. Cost reduction **54%** (**\$1.30** to **\$0.60**/query).
- Sub-agents used few-shot cards and taxonomy-driven metadata on Databricks. SQL generation precision **+50%**. Cut analysis context from **145k** to **10k** tokens.

Gameopedia | *Generative AI Engineer, Research & Engineering* Dec 2024 - Jun 2026

- Productionized **video-understanding agentic pipelines** over 3,000+ long-form videos using Gemini 2.5 Flash, CNN/ViT frame analysis, temporal scene segmentation and Python orchestration. Generated **50+ highlights/video** across games, sports, trailers, vlogs and 10+ hour streams, contributing to **+1.2% watch-time**.
- Built Game Marketing Sentiment Intelligence over **20,000+ Steam/Reddit reviews**. Established TF-IDF + Logistic Regression/SVM baselines, scaled to OpenAI Agents across 60+ categories, and deployed through **MCP + Batch API**. Cut inference cost from **\$8 to \$0.30/input** (96%).
- Fine-tuned a **360M Small-LM with LoRA** and evaluated DeepSeek-R1/GPT-3.5 across 60+ runs using loss curves, semantic similarity and qualitative review. Orchestrated taxonomy generation through LangGraph, reaching **78% semantic similarity** and **100+ new insights**.
- Architected an **RLHF-inspired autonomous refinement loop** using Gemini 2.5 Flash + OpenAI Responses API. Automated iterative evaluation and self-correction, improving classification accuracy from **33% to 88%** in three cycles.

Stealth Startup (Fintech) | *Machine Learning Engineer & AI Researcher* Aug 2024 - Nov 2024

- Audited and structured **1M+ financial prospectuses**. Built **hybrid Graph + LanceDB RAG** with chunking, embedding selection and retrieval benchmarking, then exposed it through an AWS Bedrock/Groq/LangGraph research agent for evidence-backed financial analysis.
- Built financial-document classifiers and compared embedding/ANN retrieval with fine-tuned approaches. Implemented **HYDE, CRAG and Reflexion** POCs plus an **Equity Research Agent** supporting 7+ analyst workflows with Human-in-the-Loop oversight.

Arcesium (D. E. Shaw Group) | *Forward Deployed Software Engineer, Machine Learning Practice* Mar 2023 - Aug 2024

- Built an end-to-end **tax-character prediction pipeline** using image preprocessing, OCR, feature engineering, CNNs and ensemble classifiers. Integrated it into forward-deployed enterprise processing workflows at **98% accuracy across 250+ document types**.
- Productionized an **LLM test-generation and evaluation framework** using BLEU, ROUGE, semantic similarity and statistical sampling. Compared LLaMA2, Mistral7B and CodeLLaMA, lifting test coverage from **32% to 80%** and generating **24,000 unique test cases**.
- Built a **Perl-to-Python translation evaluation pipeline**. Benchmarked 15+ LLMs on functional accuracy, AST comparison and unit-test pass rates, translating **350 functions** with Vicuna-13B at **75% functional accuracy**.
- Engineered structured/unstructured financial data and LLM pipelines using **AWS Knowledge Bases + LanceDB**. Added **Langfuse observability across 10,000+ flows** and NeMo guardrails for traceability, reliability and generation control.

SKILLS

Agentic AI & LLMs: LLMs, Agentic AI, Multi-Agent Systems, Tool Calling, RAG, Search, Ranking, Hybrid Retrieval, Rerankers, Cross-Encoders, Vector Search, Knowledge Graphs, MCP, LangGraph, LangChain, OpenAI Agents, OpenWebUI, LiteLLM, Prompt/Context Engineering.

AI Evaluation & Reliability: Retrieval and Agent Evaluation, LLM Ops, Observability, BLEU/ROUGE/F1/ROC-AUC, Human-in-the-Loop, Guardrails, Langfuse, Structured Outputs, Model Routing, Cost/Latency Optimization.

ML & Deep Learning: PyTorch, TensorFlow, HuggingFace, Scikit-Learn, Transformers, CNNs, ViTs, Temporal Models, LoRA/PEFT, Fine-Tuning, Classification, Regression, Clustering, DBSCAN, HDBSCAN, Recommendations, Ranking, Feature Engineering, Statistical Modeling, A/B Testing.

Engineering & Data: Python, FastAPI, SQL, Databricks, PostgreSQL, Redis, Qdrant, AWS Bedrock, AWS Knowledge Bases, Vertex AI, Docker, Linux, Git, React, Next.js, Vercel.

Multimodal / Media: Runway, Kling, Seedance, Veo, fal.ai, ImageKit.io, ElevenLabs, ASR, Diarization, Image/Video Generation.

PROJECTS

Fiction.ai: Multi-agent filmmaking from screenplay to shots (Director, Writer, Casting, Art, Cinematography, Editing). LangGraph with **fal.ai**, Runway, Kling, Seedance, Veo, **ImageKit.io** and **ElevenLabs**; director critique loop.

FanPulse: AI fandom intelligence for film, games and TV. Local LLM extraction over **138k** posts across **108** subreddits surfaces industry needs, demand themes and franchise risks; **HDBSCAN** clustering into **917** scored insights with grounded quotes and opportunity/risk briefs.

LifeOS: AI-native personal OS: missions, goals, projects, tasks, habits and analytics on a 90-day mission. **Next.js + PostgreSQL/Prisma** (Server Actions, JWT, Vercel). Gateway assistant with propose/apply/undo, audit ledger and skill packs.

Research Gap Finder: **9-node LangGraph** literature agent over Semantic Scholar, arXiv and OpenAlex. FastAPI + SSE + React, citation-graph analysis, BERTopic. Surfaces **7-15+** gaps with BibTeX/RIS/DOCX export.

Audio Intelligence Studio: **16-stage** FastAPI + Next.js pipeline (VAD, faster-whisper, pyannote, BERTopic, FAISS RAG). GPU-aware cost tracking. Diarization **50% faster**.

EDUCATION

UC Berkeley - Plaksha University	Post Graduate Masters, Applied AI & Product	CGPA: 8.5	2022 - 2023
KIIT	BTech, Computer Science & Engineering	CGPA: 8.01	2018 - 2022

PUBLICATIONS

Springer: “An Analytical Perspective of Machine Learning applied to Cybersecurity” | “Understanding CNN based models applied to interpreting ECG data”